Daily Algebra Drill #6

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Comparing coefficients · Substitution $t = \sqrt{x}$ for surd equations · Schur's inequality · Integer-valued polynomials.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

No expansion. Recognise the pattern, write the answer.

1. Factorise: $x^3 + 5x^2 + 6x$.

2. Given $x + \dfrac{1}{x} = \sqrt{5}$, find $x^2 + \dfrac{1}{x^2}$.

3. Without expanding, state the coefficient of xy^2 in $(x + y)^3$.

4. Evaluate: $\dfrac{1}{\sqrt{2} - 1} - \dfrac{1}{\sqrt{2} + 1}$.

5. The polynomial $f(x) = x^3 + ax + b$ has $x = 2$ as a root and $f'(2) = 0$... but no calculus today! Instead: if $x = 2$ is a *repeated* root, find a and b.

6. Simplify: $\left(\dfrac{2x}{y}\right)^3 \div \dfrac{4x^2}{y^2}$.

7. Given $\alpha\beta = 6$ and $\alpha + \beta = 5$, find $\alpha^2\beta + \alpha\beta^2$.

8. Factorise over $\mathbb{Z}$: $4a^2 - 4ab + b^2$.

9. Show quickly that $2^{10} = 1024$, hence simplify $2^{10} - 1$ as a product of three integer factors greater than 1.

10. Given $f(x) = x^2 - 3x + 2$, find $f(f(0))$.

Section B Manipulation Drills (≤50 s each)

Fractions, structure, identities. Minimum working.

1. Find constants A, B, C such that $x^2 + 3x + 5 \equiv A(x + 1)^2 + B(x + 1) + C$.

2. The polynomial $p(x) = x^3 + ax^2 + bx + c$ satisfies $p(1) = 0$, $p(-1) = 4$, and $p(2) = 15$. Find a, b, c.

3. Simplify: $\dfrac{x^3 - 8}{x^2 + 2x + 4}$.

4. Factorise: $a^3 + b^3 + c^3 - 3abc$. (*State the identity from memory, then verify by choosing* $a = b = c$.)

5. Given $a + b + c = 0$, simplify $\dfrac{a^3 + b^3 + c^3}{abc}$.

6. Prove that $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ for $n = 1, 2, 3$ by direct verification, then use the formula to find $\displaystyle\sum_{k=1}^{10} (2k-1)^2$.

7. Simplify: $\dfrac{\dfrac{1}{a+b} - \dfrac{1}{a-b}}{\dfrac{1}{a+b} + \dfrac{1}{a-b}}$.

8. Make b the subject: $\dfrac{a-b}{a+b} = \dfrac{c-d}{c+d}$.

9. Given α, β, γ are roots of $x^3 - 7x + 6 = 0$, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$.

10. Solve: $\dfrac{2}{x^2 - 1} - \dfrac{1}{x - 1} = \dfrac{3}{x + 1}$.

Section C Substitution & Structure (≤90 s each)

One clean substitution is worth ten lines of algebra.

1. Solve: $\sqrt{x} - \dfrac{6}{\sqrt{x}} = 1$. (*Let* $t = \sqrt{x}$.)

2. Find the minimum value of $x^2 - 6x + y^2 + 4y + 20$ by completing the square in both x and y.

3. Show that $a^4 + b^4 \geq \frac{1}{2}(a+b)^4$ is **false**, then find and prove the correct inequality relating $a^4 + b^4$ to $(a^2 + b^2)^2$.

4. Solve: $x^4 - 2x^3 - 3x^2 + 4x + 4 = 0$ given that it has a repeated real root.

5. Given $a, b, c > 0$ with $a + b + c = 1$, prove that $ab + bc + ca \leq \dfrac{1}{3}$.

6. If $x + y = 1$, find the maximum value of $x^2y + xy^2$.

7. Solve over $\mathbb{R}$: $\sqrt{3x+1} - \sqrt{x+4} = 1$.

8. The equation $x^2 + bx + c = 0$ has roots r and s. For what values of b and c are the roots of $x^2 + cx + b = 0$ also r and s?

Section D Challenge (SMC / BMO1 difficulty)

No brute force. Each question rewards one key observation.

1. Schur's Inequality (degree 2 case). Prove that for all non-negative reals a, b, c:

$$a^2(a-b)(a-c) + b^2(b-a)(b-c) + c^2(c-a)(c-b) \;\geq\; 0.$$

(*Consider the case $a \geq b \geq c \geq 0$ and group terms strategically.*)

2. Find all polynomials $p(x)$ with real coefficients satisfying $p(x^2) = p(x)^2$ for all real x.

3. Evaluate:

$$\frac{1^2}{1\cdot 3} + \frac{2^2}{3\cdot 5} + \frac{3^2}{5\cdot 7} + \cdots + \frac{n^2}{(2n-1)(2n+1)}.$$

Express your answer as a closed form in n.

4. A polynomial $p(x)$ of degree 3 satisfies $p(n) = \dfrac{1}{n}$ for $n = 1, 2, 3, 4$. Find $p(5)$.

5. Find all pairs of positive integers (m, n) with $m > n$ such that $m^2 - n^2 = 105$.

"In mathematics the art of proposing a question must be held of higher value than solving it." — Georg Cantor

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other prep to look at today:
Daily Integral for Calculus and Logic